

Nalish Jain

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EDUCATION

- **Indraprastha Institute of Information Technology Delhi** Delhi, India
Bachelor of Technology - Computer Science and Artificial Intelligence; CGPA: 9.8 Dec 2021 - May 2025
- **DAV Public School, Sreshtha Vihar** Delhi, India
CBSE 12th Board; Percentage: 98 Apr 2019 - Mar 2021
- **Bal Bharati Public School, Brij Vihar** Uttar Pradesh, India
CBSE 10th Board; Percentage: 97.6 Apr 2007 - Mar 2019

SKILLS

- **Languages:** Python, C++, C, Java, SQL
- **Tools & Frameworks:** PyTorch, TensorFlow, Keras, Numpy, Pandas, Scikit, NLTK, MySQL, Git, LibGDX, VSCode, IntelliJ IDEA, Figma
- **Courses:** Analysis and Design of Algorithms, Data Structures and Algorithms, Advanced Programming in Java, Operating Systems, Database Management Systems, Statistical Inference, Convex Optimization, Statistical Machine Learning, Machine Learning, Deep Learning, Reinforcement Learning, Meta Learning, Explainable AI, Artificial Intelligence, Natural Language Processing, Information Retrieval

EXPERIENCE

- **Google**
SWE Intern May 2024 - Aug 2024
 - **Project Focus:** Gemini fine-tuning for auto-intent prediction.
 - **Contribution:** Built and refined pipelines to create high-quality multilingual training datasets, then fine-tuned the Large Language models and conducted several quality iterations.
- **Signal Processing and Biomedical Imaging Lab**
Undergraduate Researcher Aug 2023 - May 2024
 - **Project Focus:** Disease classification of ECG waves using deep learning architectures.
 - **Contribution:** Developed a novel deep learning architecture using a combination of CNN and attention mechanism, resulting in significant performance improvements over state of the art results, also explored **semi-supervised learning** techniques like fix-match & mix-match.
- **Applied Solar Technologies**
Machine Learning Intern May 2023 - Jul 2023
 - **Project Focus:** Time series forecasting and anomaly detection of solar panel data.
 - **Contribution:** Using deep learning techniques like transformers, LSTM, and RNN, developed models that accurately predicted solar panel performance and detected anomalies.

PROJECTS

- **ARC-AGI Challenge** Aug 2024 - Nov 2024
LLaMA 3.2, Python, PyTorch, Meta Learning Report, Github
 - Developed and evaluated AI models on the ARC-AGI benchmark, focusing on generalization and reasoning capabilities. **Fine-tuned LLaMA 3.2 8B** to improve performance on abstract reasoning tasks. Experimented with **test-time training and test-time fine-tuning** to dynamically adapt models to novel problem instances.
- **NumerAI Challenge** Aug 2024 - Nov 2024
Python, PyTorch, Meta Learning Report, Github
 - Developed **meta learning models** for the NumerAI data science tournament using anonymized stock market data, addressing **covariate shift and concept drift** by formulating the task as a meta-learning problem. Designed and implemented models to generalize across weekly distributional changes and optimized strategies for robust stock market predictions.
- **Retrieval** Jan 2024 - May 2024
GPT-3.5, Langchain, Python, Pinecone, RAG, Information Retrieval Report, Github
 - Created a web-copilot Chrome extension enabling multimodal retrieval across multiple open tabs using Pinecone as a vector database and **GPT-3.5** for index retrieval and output generation. Integrated a chatbot interface, **RAG pipeline**, and real-time web scraping to enhance user experience, providing quick and context-sensitive responses.

- Optimizing Campus Dining** Aug 2023 - Dec 2023
Python, TensorFlow, Scikit-Learn, Machine Learning *Report, Github*
 - Developed and evaluated machine learning models using a handcrafted dataset to forecast meal sales at IIT Delhi's dining facilities. Emphasized the importance of feature engineering and encoding techniques, particularly one-hot encoding for menu items, to **optimize resource allocation and minimize food wastage** on campus.
- Replication of Dormant Neuron Phenomenon in Flappy Bird Environment** Nov 2023 - Dec 2023
Python, PyTorch, Reinforcement Learning *Github*
 - Implemented deep Q learning technique to replicate the dormant neuron phenomenon observed in the Flappy Bird environment, **achieving results similar** to those in the original paper.
- Tank Wars** Nov 2022 - Dec 2022
Java, LibGDX, Box2D *Github*
 - Developed a fast moving GUI based clone of the game "Tanks Stars" with customized tanks, weapons and live background in Java using principles of OOPs with the additional functionality of **saving games and loading saved games**.

CERTIFICATION

- Machine Learning Operations Specialization, DeepLearning.AI *Certificate Link*

HONORS AND AWARDS

- Distinguished Dean's Award of Academic Excellence 2024 for 9.5+ CGPA.
- Dean's Award of Academic Excellence 2022, 2023 and 2024 for 9+ CGPA.
- Current **rank 1** in the batch.
- JEE Advanced AIR: **2365** and JEE Main percentile: **99.86**.
- Ranked 173th among 6000+ teams in Amazon ML Challenge '23.

POSITIONS

- President**, Student Senate *Jul 2024 - Present*
- Organising Committee**, Kohinoor of Memories *Oct 2024 - Dec-2024*
- DAC Representative**, Student Senate *May 2023 - Jun 2024*
- Organiser**, Preparation for Interview Program *Jan 2024 - Jul 2024*
- Founding Member and Mentor**, Undergraduate Research Club *Sep 2023 - Jul 2024*
- Events OC**, RIISE *Jan 2024 - Mar 2024*
- Core Member**, Training and Placement Cell *Apr 2023 - Mar 2024*
- Batch Representative**, Student Council *Jun 2022 - May 2023*